Iron and Carbon: promising activity and stability in ammonia cracking

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Ammonia reforming is a crucial step in using ammonia as a hydrogen vector, but its industrial-scale implementation is hindered by the lack of efficient catalysts. Ruthenium on carbon is a promising catalyst, but cheaper alternatives like iron, nickel, and cobalt are also showing promising activities. However, carbon-supported catalysts have a major drawback: they tend to undergo methanation at high temperatures in hydrogen atmospheres.[1-3]

We synthesized and characterized a series of catalysts consisting of nickel, iron, and cobalt incorporated in carbon materials. The catalytic activity in ammonia reforming was tested in the temperature range of 400-600 °C, and the carbon stability was investigated using thermogravimetry (TG) coupled with mass spectrometry.

Our results show that carbon-supported Fe presents promising activity in ammonia reforming, reaching a conversion rate of ~10 $kg_{NH3}/kg_{catalyst}h$ at 550 °C, which is close to commercial bulk Iron catalysts. Within long-term screening experiments, no deactivation of the catalysts was observed during 500 h of testing at up to 600 °C. At ≥ 600 °C continuous mass loss is observed in TG-experiments. This mass loss is low when using Fe as active species while high when using Ni or Co containing catalysts. In contrast, reversible nitridation/de-nitridation is observed in a limited to a short time interval during ramping up/down of temperature in TG. The clear separation of nitridation and degradation make the TG-experiments to a very informative tool for investigation of processes during ammonia reforming (see Figure 1). Our study demonstrates that Fe supported on carbon is a highly promising catalyst for ammonia cracking, offering both high activity and stability. Notably, Fe/C catalysts exhibit superior stability compared to other earth-abundant carbon-supported transition metal catalysts, making them an attractive option for industrial-scale ammonia reforming applications.

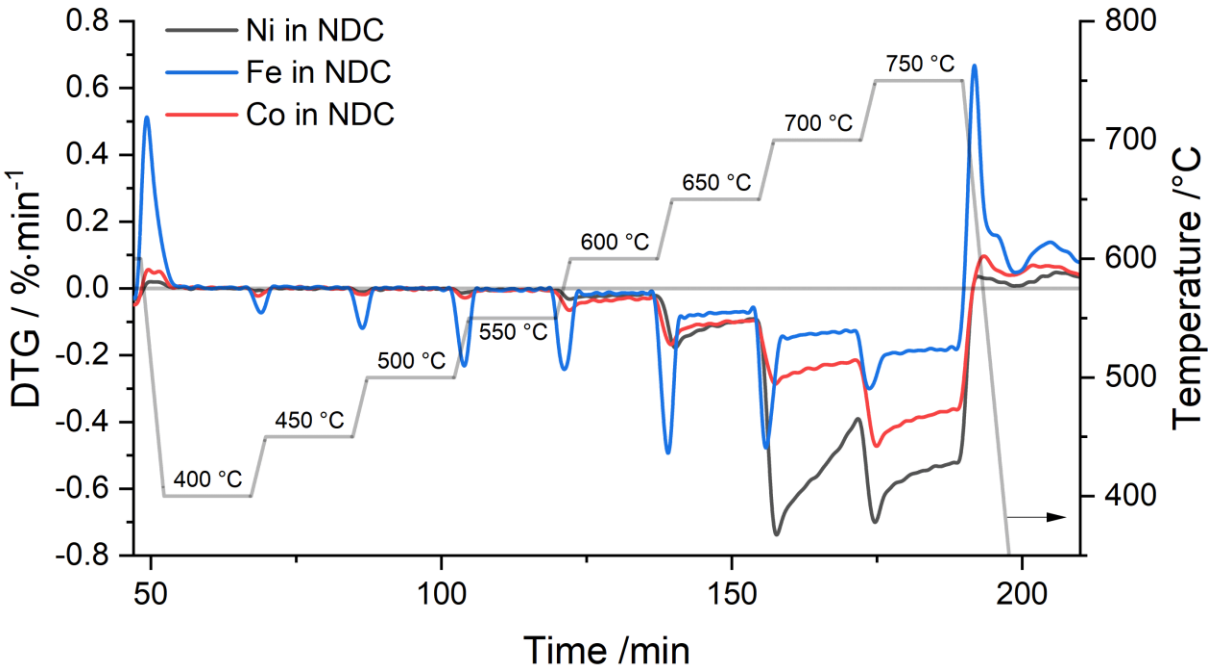

Figure 1: 1st derivative of catalyst mass over time in 30% NH_3:He atmosphere of Fe, Co or Ni carbon-supported monometallic catalysts obtained from TG-MS experiments showing the instability of such catalysts at > 600 °C.

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